

Here's an example that uses stateless widgets:

```
class Demo extends StatelessWidget {
  const Demo({ Key key }) : super(key:
key);

  @override
  Widget build(BuildContext context) {
    return new Column();
  }
}
```

On the other hand, stateful widgets have mutable states. So when does one use them? They can be used when the UI is prone to dynamic changes.

Here's an example that uses stateful widgets:

```
class Demo extends StatefulWidget {
  const Demo({ Key key }) : super(key:
key);

  @override
  _DemoState createState() => new _
DemoState();
}

class _DemoState extends State<Demo> {
  @override
  Widget build(BuildContext context) {
    return new Column();
  }
}
```

Another type of widget is used when we have widgets subscribed to a single widget, and we want to efficiently pass on information to all the subscribed widgets when the single widget changes its state. That is, when an inherited widget changes, all the subscribed widgets will be re-drawn.

Learn more about inherited widgets at <https://docs.flutter.io/flutter/widgets/InheritedWidget-class.html>.

Gestures

Gestures can be in the form of a tap, a double tap, a long press, vertical and horizontal drags, etc.

Flutter provides us with the GestureDetector widget. For example:

```
new GestureDetector(
  onTap: () {
    setState(() { _birthday = true});
  },
  child: new Container(
    child: new Text('Happy Birthday!'),
  ),
)
```

Flutter has plenty of features of its own, and can also be used for developing many interesting things such as animations and introducing internationalisation. Head over to flutter.io to develop your first Flutter app!

The best thing about Flutter is that developers from any platform can relate to Flutter right away! Many of Flutter's components can be related to the components used presently in the Web, Android as well as iOS. As a result, it becomes easy for developers to relate their knowledge to Flutter.

Flutter for Web developers

Here is an example of HTML code used in Flutter for the Web:

```
.text{
  font: 200 14px Arial;
}
```

An example of Dart code is given below:

```
Text(
  "Happy Birthday",
  style: new TextStyle(
    fontSize: 14.0
    fontWeight: FontWeight.w200,
    fontFamily: "Arial"
  ),
)
```

Another example of HTML code is:

```
.contain{
  background-color: #26c6da;
}
```

Another example of Dart code is:

```
Container(
  color: Colors.cyan[400]
)
```

Flutter for Android developers

As an Android developer, you can easily apply your knowledge to build mobile apps with Flutter. It provides many components that are equivalent to the components in Android. So it becomes a lot easier to visualise your app in Flutter.

1. Android Views are roughly equivalent to widgets in Flutter: *View* is the foundation of everything in Android and in Flutter, everything is a widget. But widgets are not strictly Views, as they are descriptions of the UI that gets converted into the actual View objects.
2. Material library: Flutter uses widgets that implement the material design guidelines which are also used in Android.
3. Draw/Paint: In Android, we use *Canvas* and *Drawable*. Flutter uses a similar *Canvas* API. So, for Android developers, these are familiar tasks in Flutter.
4. Navigation: Android uses *Intents* to navigate between *Activities* and Flutter uses *Navigator* and *Routes* to navigate between screens.

Head over to the official website of Flutter at <https://flutter.io/flutter-for-android/> to find out the various other similarities and differences with Android.

Flutter for iOS developers

As an iOS developer, you can easily apply your knowledge to build mobile apps with Flutter, which provides many components equivalent to those in the iOS framework. So it becomes a lot easier to visualise your app in Flutter.

1. iOS *UIView* is roughly equivalent to widgets in Flutter: *UIView* is the foundation of everything in iOS and in Flutter, everything is a widget. But widgets are not strictly UIViews, as they are a description of the UI that gets converted into the actual *UIView* objects.
2. Material library: Flutter uses widgets that implement the material design guidelines, which are optimised for iOS too.

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